

ResponSight: Explainable and Collaborative Moderation Approach for Responsible Video Content in UGC Platform

Haonan Nan
Antai College of Economics and
Management
Shanghai Jiao Tong University
Shanghai, China
nanhaonan@sjtu.edu.cn

Zixiao Wang
Human Centered Design &
Engineering
University of Washington
Seattle, Washington, USA
wzxwzx294@163.com

Yu Zhang
Antai College of Economics and
Management
Shanghai Jiao Tong University
Shanghai, China
zhangyu_candy@sjtu.edu.cn

Xuantao Zhang
Antai College of Economics and
Management
Shanghai Jiao Tong University
Shanghai, China
dickytao@sjtu.edu.cn

Saixing Zeng
Antai College of Economics and
Management
Shanghai Jiao Tong University
Shanghai, China
zengsaixing@sjtu.edu.cn

Abstract

The rapid spread of user-generated video content on social media has raised pressing ethical concerns, requiring a holistic approach to moderate the contents. While current research primarily focuses on explicit text-content violations detection, it overlooks the ethical dimensions of content responsibility and users' needs for clear explanations and specific guidance. In response, we construct a novel video moderation system for user-generated content (UGC) platforms, shifting from traditional reactive approaches to a proactive, distributed human-machine workflow. We design a detection tool framework called *ResponSight* that collaborates with users to promote responsible content creation. *ResponSight* contains two core modules: the explainable evaluation module and the adaptive suggestion module, which driven by multimodal large language models (MLLMs). By offering transparent, explainable feedback, this system enables users to understand the rationale behind moderation decisions and refine their videos to align with ethical and social responsibility standards.

CCS Concepts

• Human-centered computing; • Collaborative and social computing; Collaborative and social computing systems and tools.;

Keywords

responsible video content, explainable moderation, human-AI collaboration, multimodal large language model

ACM Reference Format:

Haonan Nan, Zixiao Wang, Yu Zhang, Xuantao Zhang, and Saixing Zeng. 2025. ResponSight: Explainable and Collaborative Moderation Approach for Responsible Video Content in UGC Platform. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3706599.3720230>

1 Introduction

Social media platforms have become essential spaces for interactive communication, enabling users to share and access information instantly [1]. These platforms are primarily driven by user-generated content (UGC), where users play dual roles as both creators and consumers of diverse content forms [2]. The platform reduced barriers to video publication, leading to increased participation by non-professional users and a corresponding decline in content quality control [3]. The irresponsible content can mislead public perception, intensify social polarization, and negatively impact vulnerable groups through misinformation [4], discriminatory messages [5], and cyberbullying or exploitation content [6, 7]. Given these risks, both platforms and users have a growing responsibility to ensure that content adheres to responsible creation standards. Responsible video content is defined as content that, beyond adhering to legal regulations and platform policies [8], upholds ethical standards and social responsibility [9], by considering its potential impact on society [10]. Creating responsible video content represents an initiative aimed at encouraging creators to not only follow basic regulations but also actively contribute to societal well-being, promote fairness, support diversity, prevent harm, and limit the spread or manipulation of negative emotions [11–13].

Video content moderation is complex due to its multimodal nature, involving text, audio, and visual elements [14]. Current moderation systems often rely on a hybrid approach, combining automated detection tools with human moderator oversight, which typically follows a reactive, two-stage process [15]. However, existing research primarily focuses on text-based content moderation, with limited attention given to video content, which presents unique challenges. Current methods, such as rule-based algorithms and

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '25, Yokohama, Japan
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1395-8/2025/04
<https://doi.org/10.1145/3706599.3720230>

machine learning models [16–18], often suffer from high error rates [19], lack of transparency [20], and inadequate handling of context-sensitive issues [21]. Furthermore, these systems fail to address the specific needs of responsible video moderation, particularly in terms of ensuring ethical content creation and improving user engagement [9, 22]. There is a gap in the literature regarding the development of automated systems for responsible video moderation, with insufficient focus on how to design and implement such systems. Therefore, there is a clear need to propose a new moderation system that not only improves efficiency and accuracy but also emphasizes transparency, fairness, and responsibility in video content moderation. We focus on two main research questions: 1) How to design a collaborative system for UGC platform content moderation; 2) How to build an explainable and adaptive framework of detection tool for moderation to nurture more responsible videos.

Building a moderation system for video content moderation faces several significant challenges. First, improving efficiency through automated tools while reducing manual labor remains a critical issue. Although machine learning and rule-based algorithms can process large volumes of content quickly, they often suffer from inaccuracies such as false positives and negatives, which necessitate human intervention. This imbalance leads to inefficiencies and increases operational costs, making the automation of the moderation process an ongoing challenge. Second, it is necessary to promote responsible content creation among users. While current systems largely rely on punitive measures, they fail to encourage proactive, responsible content production. Developing effective intervention strategies that motivate users to self-regulate and reduce the creation of harmful content by granting users sufficient autonomy. Finally, ensuring the explainability of automated moderation results is vital for enhancing user trust and engagement. Automated systems often lack transparency, leaving users frustrated when they don't understand why their content is flagged or removed. Therefore, integrating techniques that can provide explainable results and clear, actionable insights into moderation decisions could significantly improve user experience, making the moderation process more understandable and fairer.

For the above problem and challenges, we propose a conceptual framework of a responsible video moderation system and design an intelligent tool framework for moderation. First, through literature analysis, we identify the limitations of existing moderation systems and methods, which inform the design direction of our system. We address the challenges of traditional moderation workflows, where users, algorithms, and moderators operate in fragmented, sequential stages, often leading to delays and low transparency. Second, to overcome this, we propose a distributed self-moderation system, incorporating proactive identification of key responsible video dimensions, designing a novel moderation pipeline and introducing a human-AI collaboration model to enhance the efficiency, transparency, and responsiveness of UGC content moderation. This system enables users to self-assess and refine their videos, improving both content quality and the overall user experience. Third, we design and implement the *ResponSight* framework, a transparent and adaptive video detection tool driven by multimodal large language models (MLLMs). This framework analyzes user-generated videos by leveraging MLLM's capabilities in summarization and

semantic understanding to detect content issues effectively. It provides real-time feedback and actionable modification suggestions, tailored to different video types through a personalized prompt generator. Finally, we conduct preliminary validation, offering insights into the advantages of an explainable self-moderation system from both user and platform perspectives. Our contributions are threefold:

- Constructing a novel video moderation system for UGC platforms, transforming the traditional reactive process into a distributed, proactive, and integrated human-machine workflow.
- Designing an intelligent detection tool framework that collaborates with users to support responsible video creation, empowering users with actionable guidance for refining their content to align with ethical and social responsibility standards.
- Proposing several insights on how responsible video moderation fosters a healthier and more trustworthy digital ecosystem for UGC communities.

2 RELATED WORK

Content moderation in online communities has been a rich research area in HCI, with existing studies including workflows that combine automation and manual review [15, 23, 24], the importance of transparency and explanations in moderation [20, 23, 25], the application of various algorithms in detecting violations [20, 27–29], and the impact of content moderation on user behavior [15, 29, 30], particularly how human-computer collaboration can improve the efficiency of moderation and the user experience [30–32].

2.1 Content Moderation System in UGC Platform

Content moderation systems across UGC platforms typically rely on a blend of automated systems and human oversight to detect and manage harmful content. These systems often use machine learning algorithms for automated detection, while human moderators intervene in more complex cases [25, 33, 34]. This process begins with automated systems that use machine learning and rule-based algorithms to detect potential violations such as explicit or harmful content [35], followed by human review to validate or further examine flagged content. While this model ensures basic compliance with platform standards, it presents several significant limitations that undermine its effectiveness and user engagement [17].

A primary issue with traditional moderation is its lengthy review time and heavy workload. Automated systems are fast but frequently inaccurate, leading to high rates of false positives or negatives [19]. Human moderators frequently deal with high volumes of flagged content, which can lead to cognitive overload, emotional fatigue, and burnout [36, 37]. Recent research increasingly advocates for human-machine collaboration in content moderation [23, 27, 36]. Automated systems can assist by processing large volumes of content, while human moderators can intervene when content is ambiguous or context-sensitive [23, 38]. Recent studies have shown that combining the efficiency of automation with the nuanced understanding of human judgment leads to more effective moderation systems [23, 36].

Another significant challenge is the lack of transparency and the associated negative impact on user experience. Automated tools such as Reddit’s AutoModerator have shown limitations in context-sensitive content moderation, where they fail to provide transparency in their actions [38, 39]. Users often do not receive clear explanations regarding why their content was flagged or removed, leading to dissatisfaction and mistrust [40]. The absence of such transparency exacerbates user frustration, reducing engagement and potentially undermining trust in the platform’s moderation decisions [29, 41]. Hartwig et al. designed an indicator-based intervention to address transparency needs in video-sharing platform (VSP) [42]. When an explanation for removal is provided, the willingness of users to improve their posting behavior will increase [29]. The integration of explainable AI (XAI) can significantly improve users’ understanding of moderation actions, thereby enhancing engagement and satisfaction [41, 43].

Meanwhile, current moderation practices mainly focus heavily on punitive actions, such as flagging or removing non-compliant content [44]. In this situation, users are more likely to avoid explicit violations rather than considering the social impact or ethical quality of their content. Allowing users to challenge moderation decisions or adjust flagged content based on feedback has been shown to increase both fairness and accuracy [27, 45]. The inclusion of personalized feedback and appeals mechanisms enables content creators to better understand why their videos were flagged and make appropriate adjustments [46]. This interactive process also helps refine the AI models used in moderation, ensuring they adapt to changing patterns in content and user behavior [20, 27]. Research shows that when users can influence moderation decisions, they are more likely to feel invested in the platform and engage more responsibly [47].

2.2 Video Violations Detection

Content moderation systems have traditionally focused on text-based content, recent studies use pre-trained language models, neural networks integrated attention mechanisms, and other models in misinformation, violence and other violate content detection [16–18, 35]. Compared with text-based content, video content involves multiple modalities, including visual, auditory, and textual elements, which complicate automated detection [48]. The inclusion of multimodal data could significantly improve the ability to detect violations in video content [49], but this remains underexplored in existing research. Given the dynamic nature of video content, it is important to develop more sophisticated detection methods with the ability to analyze multimodal data in real-time, ensuring that harmful content is flagged before it can cause damage.

Explainability is particularly important for video content, where the combination of multiple data sources (text, audio, visual) makes it difficult to provide clear reasoning for automated decisions [31, 43]. Providing users with explainable feedback, such as why specific segments of a video were flagged or how certain features were identified as problematic, enhances transparency and user trust in the system [29, 41]. Adaptive systems that adjust based on user behavior and feedback can also improve detection accuracy and ensure moderation is more context-sensitive [46, 50].

The integration of large models (LLMs) in content moderation has shown potential, particularly in detecting toxic language and harmful content at scale [51–53]. These models benefit from advanced capabilities such as zero-shot learning and fine-tuning on specific tasks, making them adaptable to a wide range of content moderation applications [26, 53]. However, LLMs also have limitations, including the potential for bias in their training data and challenges in interpreting the diverse and complex content [51, 53, 54].

Based on the analysis of the above studies, our work focuses on establishing a responsible video content moderation system that integrates human-AI collaborative process and explainable multimodal detection, which are directly aligned with HCI principles. The system provides transparent explanations for decisions, and allows for user involvement, which can enhance both trust and engagement with the platform [40, 43, 55]. The combination of multimodal analysis and personalized feedback also ensures that content creators are empowered to make informed decisions about their content, fostering a more responsible and ethical creation process [46, 50]. This collaborative approach is central to HCI research, which emphasizes the need for user-centered, transparent, and adaptable systems [20, 47].

3 RESPONSIBLE VIDEO MODERATION SYSTEM

3.1 Proactive Identification of Responsibility

Previous research has primarily focused on detecting specific types of harmful content, such as misinformation and toxic speech [28, 56]. These studies typically employ a reactive approach, acting against harmful content after it has been published. In contrast, a proactive nurturing approach encourages users to reconsider harmful behavior before it occurs [57]. Our research shifts the focus from reactive moderation to proactive identification. Building on previous studies, we integrate key dimensions of responsibility in content moderation and propose six core dimensions for responsible video detection: “Compliance” evaluates whether video content strictly adheres to relevant legal standards and platform guidelines, ensuring that all material avoids prohibited elements; “Reliability” focuses on the accuracy and credibility of video content, ensuring it is based on trustworthy information and free from misleading or false claims; “Fairness and Inclusivity” assesses whether the video avoids discriminatory language or biases and respects diversity; “Privacy and Data Security” reviews the video’s handling of personal information, ensuring that privacy is respected, and sensitive data is not disclosed without consent; “Social Impact and Responsibility” examines the broader societal implications of video content, ensuring it does not contribute to negative social issues or mislead public understanding; “Appropriateness of Emotional Expression” considers whether the emotional tone in the video aligns with its content, avoiding exceed manipulation that could lead to collective negative reactions. The specific connotations of each dimension and the corresponding rating standards are provided in Appendix A.1.

These dimensions equip platforms and creators to systematically identify, assess, and enhance content quality in alignment with ethical and social standards, fostering a media environment that

prioritizes safety, inclusivity and transparency, ultimately contributing to a healthier, more trustworthy digital ecosystem.

3.2 System Design Goal

The moderation system design addresses three primary goals aimed at creating a more responsive, transparent, and user-centered moderation framework. First, it emphasizes real-time responsiveness, facilitating proactive content review during both the creation and distribution phases. This means that users can engage in self-evaluation and make instant adjustments to their content in the production or upload phase without the typical delays associated with traditional “review-feedback-revision” cycles. Furthermore, the framework’s continuous assessment capabilities allow for dynamic monitoring of content impact throughout its dissemination, providing platforms with data to track and respond to the evolving influence of shared content. Further, the framework is expected to ensure explainability and adaptability within the evaluation and suggestion mechanisms. Unlike traditional systems that function as black-box solutions, the framework generates clear, actionable feedback, equipping creators with focused recommendations that align with both platform policies and broader social responsibility standards. This transparency not only aids creators in refining their content more effectively but also enhances their understanding of compliance and responsible criteria. It also focuses on user-centered human-AI collaboration, where *ResponSight* operates based on user authorization and active engagement. This approach establishes a collaborative ecosystem in which users, platforms, and the machine work together toward responsible content creation. In this ecosystem, users are positioned as central agents of content refinement, leveraging machine assistance to enhance their content responsibly, while the platform functions as the supportive infrastructure that facilitates and sustains these interactions.

3.3 Moderation Pipeline with ResponSight

Our research introduces *ResponSight*, a video detection tool framework based on MLLMs that offers explainable evaluation and adaptive suggestions for responsible content creation. Through human-AI collaboration, the framework allows users and UGC platforms to dynamically identify and improve elements that may not meet responsibility standards. Leveraging its multi-modal capabilities, the framework interprets various video modalities—such as visuals, audio, and text—to deliver actionable, context-aware feedback, enabling users to make real-time adjustments before content publication.

The framework’s adaptive nature seamlessly integrates into the video upload and review process, transforming the traditional content publication workflow (as shown in Figure 1). At the upload stage, users can activate *ResponSight* to receive immediate evaluation reports with actionable suggestions, allowing for quick adjustments to align content with platform standards. This real-time feedback loop accelerates content approval by enabling high-rated videos to be published promptly while guiding necessary revisions for compliance. Beyond the upload phase, it supports routine post-publication checks, where platform moderators can reference its detailed evaluative reports to streamline their process. By highlighting critical content risks and compliance issues, these reports reduce

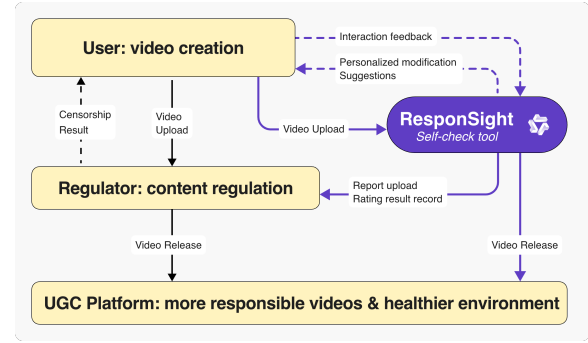


Figure 1: Proposed Video Moderation Pipeline

manual oversight, enabling reviewers to focus on high-risk content, thereby boosting moderation efficiency. Embedding *ResponSight* across the content lifecycle—from creation to post-publication—shifts the platform’s approach from reactive moderation to proactive management. Users are empowered to enhance content responsibility, the AI-driven framework serves as an assistive partner, and human moderators can prioritize oversight more strategically. Human moderators only need to focus on videos with low ratings that have not been modified or those that remain low-rated after re-uploading. They can refer to the detailed review results provided by the tool, eliminating the need to watch the entire video. Subsequent random checks can simply retrieve review records, significantly reducing the workload.

3.4 Human-AI Collaboration Model in Moderation

The model design aims to facilitate a seamless, user-centered experience that balances explainability, responsiveness, and user autonomy in video content evaluation and modification (as shown in Figure 2). When users activate *ResponSight* during video upload, the system immediately evaluates the video via the explainable evaluation module. The evaluation results are presented in a clear, visually organized report, making it easy for users to understand the video’s performance in each dimension. This feedback panel integrates concise, prioritized suggestions for improvement, enabling users to focus on the most critical areas first. Based on its real-time feedback loop feature, *ResponSight* allows users to make content adjustments and immediately receive updated evaluation results. This mechanism replaces the traditionally lengthy “evaluation-feedback-revision” cycle, significantly reducing the time between user actions and system responses. As a result, users can iteratively refine their content in a streamlined and responsive process, maintaining control over each adjustment phase without waiting for extended feedback delays. To ensure that users fully understand and trust the evaluation results, each feedback item includes a brief reason explaining why the video element requires improvement. This explainability is crucial for supporting user decision-making, as it helps demonstrate the system’s evaluation criteria, transforming assessments into actionable insights.

Moreover, the model includes an embedded feedback option, allowing users to rate the relevance of specific suggestions. By

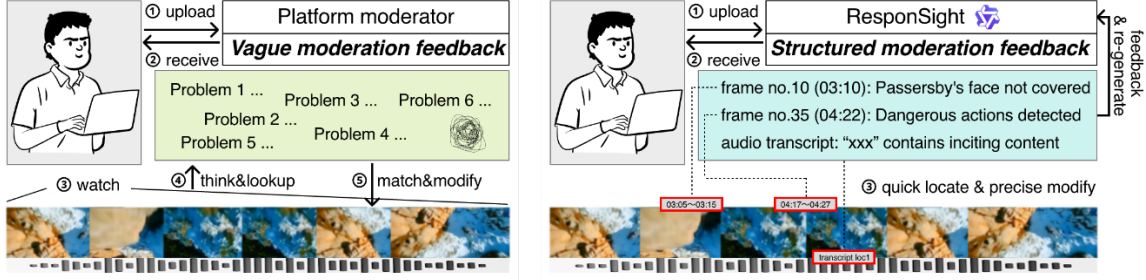


Figure 2: Collaboration Model (left: without AI tools; right: with AI tools)

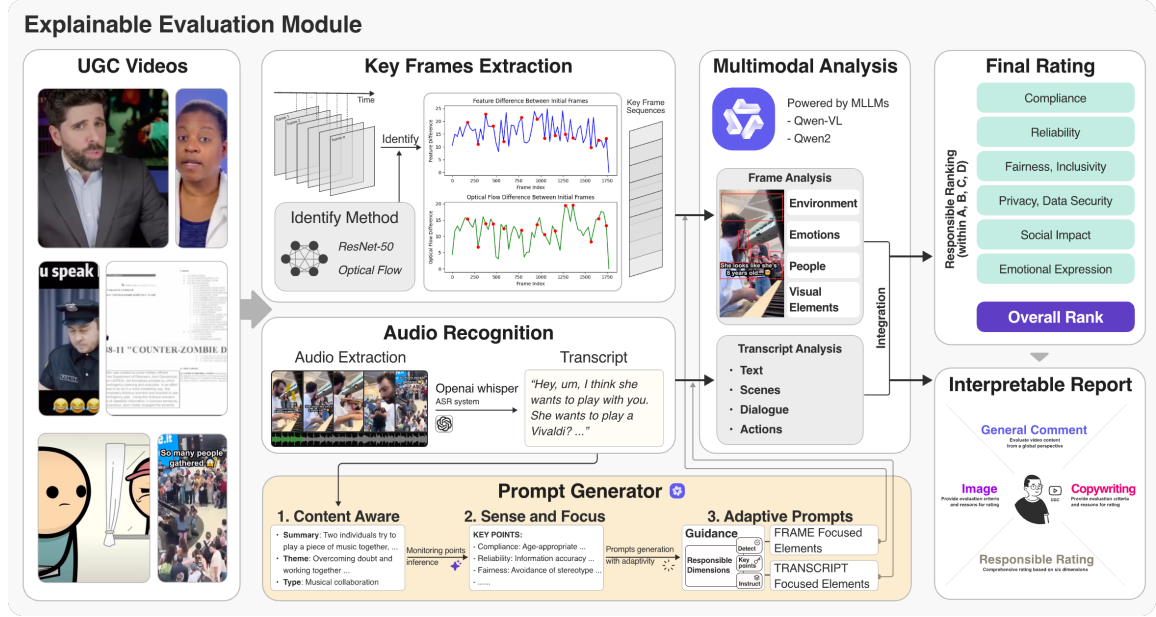


Figure 3: Explainable evaluation module structure

incorporating the mechanism of LLM “Learning from Human Preference” [58], *ResponSight* allows user feedback to be integrated back into the system’s model, fostering a continuous feedback loop that optimizes the model’s accuracy, adaptability, and alignment with user preferences. This iterative improvement process enhances the system’s intelligence, tailoring it to better meet user and platform expectations over time. This feedback is directly integrated into the model learning process, enabling the system to progressively align its recommendations with user expectations and platform standards.

4 TECHNICAL FRAMEWORK

ResponSight is centered on two core technical modules: the explainable evaluation module and the adaptive suggestion module, each fulfilling critical roles in video assessment and recommendation generation. The framework is instantiated using the Qwen series of large models, driven by the robust performance of Qwen-VL and Qwen2 [59, 60].

4.1 Explainable Evaluation Module

After a user uploads a video, the content undergoes an evaluation in the explainable evaluation module to assess its rating of responsibility. This module consists of four core components (as shown in Figure 3): video decomposer, prompt generator, MLLMs-based detector, and rating function. The specific technical components are shown in Appendix A.2.

4.2 Adaptive Suggestion Module

Following the generation of the evaluation report, the adaptive suggestion module provides creators with clear, actionable modification guidance in accordance with the report’s findings. This module is designed to help optimize content to enhance both the visibility and responsibility level of the video content. This module is powered by the LLM-based adviser (as shown in Figure 4). The specific technical components are shown in Appendix A.3.

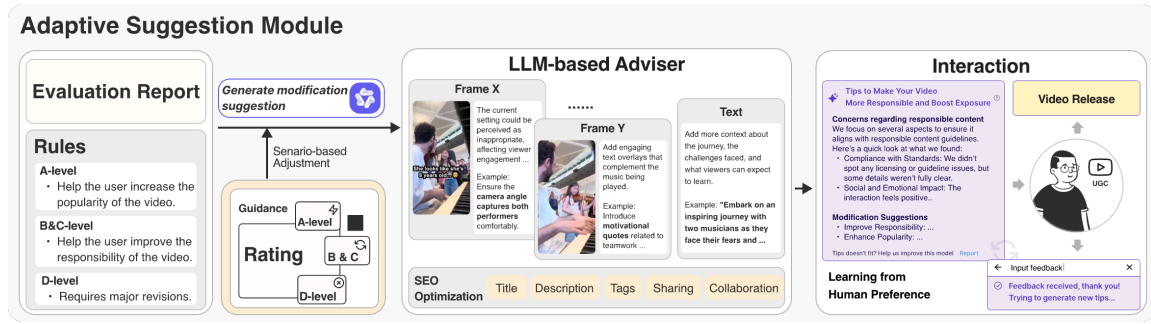


Figure 4: Adaptive suggestion module structure

5 DISCUSSION AND CONCLUSION

5.1 Implications

Our work presents a novel approach to video moderation by proposing a conceptual framework for responsible video moderation systems and designing an intelligent detection tool framework. We address the limitations of traditional moderation workflows by introducing a proactive, distributed self-moderation system, which integrates human-AI collaboration to enhance efficiency, transparency, and responsiveness. The *ResponSight* framework offers real-time feedback and actionable modification suggestions driven by MLLMs, empowering users to refine their content. Our work contributes to transforming UGC content moderation from a reactive to a proactive process, promoting responsible content creation and improving the overall user experience. These innovations lay the foundation for future research and development in user-driven, ethical content moderation systems.

Our work shifts the structural dynamics of social media platforms. Traditionally, content moderation on these platforms relied on reactive, post-upload moderation processes, where only platform administrators possessed the authority to enforce content standards. With *ResponSight*, the moderation process has been transformed to include user-centered, proactive evaluation directly at the point of content creation. Users are now able to self-assess and receive actionable guidance on their content’s responsibility level before publishing, actively participating in the quality control process. Consequently, this shift has decentralized the control over content standards from exclusive platform administrators to a shared responsibility model between users, the AI-driven evaluation system, and platform administrators. This new workflow allows for a more interactive and iterative moderation loop that aligns content with social responsibility standards.

5.2 Deployment Challenges

Given the focus of this study on content moderation practices in UGC platforms, it is crucial to discuss the practical challenges of real-world deployment. These challenges primarily revolve around computational overhead, scalability, adaptability to various cultural environments, and mitigating malicious or non-cooperative user behaviors.

One major challenge in deploying MLLMs is the computational resources required for real-time video analysis. These models demand substantial computational power, especially when processing large volumes of video data. Balancing efficiency and accuracy is key for ensuring the system can handle high traffic without sacrificing moderation quality. A potential solution is distributed computing, where tasks are decomposed and assigned to different devices, ensuring that the computational load is efficiently distributed. However, managing data queues and ensuring smooth integration across these devices is a critical challenge, requiring robust data communication and orchestration mechanisms. From a deployment perspective, edge-cloud collaboration offers a promising solution. The core models deployed in the cloud can process video content to extract metadata, then utilizing the processing capabilities of edge devices (user terminals) to provide localized, smaller-scale inferences, the system can reduce the strain on central servers. This strategy could significantly reduce the computational burden on the platform’s centralized systems, allowing for greater scalability without compromising user experience or system performance.

Another challenge is adapting *ResponSight* to different platforms and cultural contexts. Content moderation is subjective, and ethical standards vary across cultures. To address this, the model should be pre-trained on platform-specific video data, with historical moderation datasets used for fine-tuning. System prompts can also be adjusted to guide moderation according to local norms, ensuring that the system aligns with diverse regulations.

A significant concern is the potential for malicious or non-cooperative users to provide misleading feedback to manipulate the system. In a user-driven moderation model, bad actors could label valid alerts as false positives, degrading the system’s accuracy. We suppose a multi-layered isolation mechanism could mitigate this. The first layer separates the user interface from the backend system, ensuring feedback is verified before affecting the model. The second layer isolates the data communication channel from the core model, preventing malicious inputs from contaminating the model. Additionally, a warning mechanism will be beneficial, including lowering the model’s sensitivity to low-confidence feedback and automatic re-evaluation for the flagged videos before final judgment.

5.3 Limitations and Future Work

This work is currently in the early stages of technological innovation, with a focus on developing a prototype and testing its feasibility. Due to the performance limitations of locally deployed large models, a full-scale performance validation of the proposed framework has not yet been completed. For now, we conducted a preliminary study by collecting a dataset of 1,453 videos from platforms such as *YouTube*, *X*, *TikTok*, and *Bilibili*. From this dataset, a random sample of 50 videos was selected and evaluated using the *ResponSight* framework. The initial results from manual evaluations of the framework's output suggest that this human-AI collaboration approach is effective in rating video content, generating explainable assessments, and providing actionable suggestions for improvement. We present two testing examples in appendices A.4. The small sample size in preliminary experiments may lead to results that are not fully representative, and there has been no dedicated evaluation dataset constructed for comprehensive testing. Additionally, while existing research has explored various aspects of content moderation, there is still a lack of established evaluation systems for video responsibility assessment. This gap highlights a promising area for future research to address.

To ensure the framework's robustness and practical relevance, future research will adopt a comprehensive, multi-faceted validation strategy. Initially, performance evaluation will involve quantitative metrics based on large model evaluation standards to assess the overall effectiveness of the system. Following this, a series of user-centered experiments will be conducted to examine how users perceive and accept the modification suggestions provided by the system. This phase will yield valuable insights into the practical usability and user acceptance of these recommendations in the content creation process. To assess the real-world applicability of *ResponSight*, we plan to deploy testing in selected platforms, comparing the system's impact on user interactions and content quality. This approach will allow us to gauge the framework's effectiveness in enhancing video quality, ensuring social responsibility standards, and improving the efficiency of video moderation processes. Moreover, future work will focus on user needs assessments, usability testing, and the iterative refinement of functionality, interaction design, and system deployment. This will involve understanding the requirements of both users and moderators, adjusting the system to provide meaningful feedback, and ensuring that the user experience aligns with practical content creation and moderation needs.

Acknowledgments

We would like to express our gratitude to all the participants who provided insightful suggestions and interesting ideas for our research. We also thank the ACs and the reviewers for their valuable comments that significantly enhanced this paper. This work is supported by the National Natural Science Foundation of China (Grant no: 72394375).

References

- [1] Tracy L. Tuten. 2008. *Advertising 2.0: Social Media Marketing in a Web 2.0 World*. Praeger Publishers, USA.
- [2] Matthew S. Eastin Terry Daugherty and Laura Bright. 2008. Exploring Consumer Motivations for Creating User-Generated Content. *J. Interact. Advert.* 8, 2 (2008), 16–25. <https://doi.org/10.1080/15252019.2008.10722139>
- [3] Wei Zhuang, Qingfeng Zeng, Yu Zhang, Chunmei Liu, and Weiguo Fan. 2023. What makes user-generated content more helpful on social media platforms? Insights from creator interactivity perspective. *Inf. Process. Manag.* 60, 2 (2023), 1–14. <https://doi.org/10.1016/j.ipm.2022.103201>
- [4] Jinghua Piao, Jiazhen Liu, Fang Zhang, Jun Su, and Yong Li. 2023. Human-AI adaptive dynamics drives the emergence of information cocoons. *Nat. Mach. Intell.* 5, 11 (2023), 1214–1224. <https://doi.org/10.1038/s42256-023-00731-4>
- [5] Xiangyu Tao and Celia B Fisher. 2022. Exposure to social media racial discrimination and mental health among adolescents of color. *J. Youth Adolesc.* 51, 1 (2022), 30–44. <https://doi.org/10.1007/s10964-021-01514-z>
- [6] Zainab Agha, Kelsey Miu, Sophia Piper, Jinkyung Park, and Pamela J. Wisniewski. 2023. Co-Designing User Personas and Risk Scenarios for Evaluating Adolescent Online Safety Interventions. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing* (Minneapolis, MN, USA) (CSCW '23 Companion). Association for Computing Machinery, New York, NY, USA, 249–253. <https://doi.org/10.1145/3584931.3606964>
- [7] Guofang Nan, Ning Ding, Guangyu Li, Zhiyong Li, and Dahui Li. 2023. Two-tier regulation models for the user-generated content platform: A game theoretic analysis. *Decis. Support Syst.* 175 (2023), 1–17. <https://doi.org/10.1016/j.dss.2023.114034>
- [8] Renkai Ma and Yubo Kou. 2021. "How Advertiser-Friendly is My Video?": YouTube's Socioeconomic Interactions with Algorithmic Content Moderation. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 429 (oct 2021), 25 pages. <https://doi.org/10.1145/3479573>
- [9] Kristen Vaccaro, Ziang Xiao, Kevin Hamilton, and Karrie Karahalios. 2021. Contestability For Content Moderation. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 318 (oct 2021), 28 pages. <https://doi.org/10.1145/3476059>
- [10] Farhana Shahid. 2024. Human-AI Collaboration to Facilitate Culturally-Aware Content Moderation. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing* (CSCW Companion '24). Association for Computing Machinery, New York, NY, USA, 9–12. <https://doi.org/10.1145/3678884.3682057>
- [11] Jesse Graham, Brian A Nosek, Jonathan Haidt, Ravi Iyer, Spassena Koleva, and Peter H Ditto. 2011. Mapping the moral domain. *J. Pers. Soc. Psychol.* 101, 2 (2011), 366–385. <https://doi.org/10.1037/a0021847>
- [12] Zeeshan Ahmed Bhatti, Ghulam Ali Arain, Muhammad Shakaib Akram, Yu-Hui Fang, and Hina Mahboob Yasin. 2020. Constructive voice behavior for social change on social networking sites: A reflection of moral identity. *Technol. Forecast. Soc. Change* 157 (2020), 1–14. <https://doi.org/10.1016/j.techfore.2020.120101>
- [13] Dominik Neumann and Nancy Rhodes. 2024. Morality in social media: A scoping review. *New Media Soc.* 26, 2 (2024), 1096–1126. <https://doi.org/10.1177/1461448231166056>
- [14] Han Wang, Tan Rui Yang, Usman Naseem, and Roy Ka-Wei Lee. 2024. Multi-HateClip: A Multilingual Benchmark Dataset for Hateful Video Detection on YouTube and Bilibili. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*. Association for Computing Machinery, New York, NY, USA, 7493–7502. <https://doi.org/10.1145/3664647.3681521>
- [15] Jialun Aaron Jiang, Peipei Nie, Jed R. Brubaker, and Casey Fiesler. 2023. A Trade-off-centered Framework of Content Moderation. *ACM Trans. Comput.-Hum. Interact.* 30, 1, Article 3 (February 2023), 34 pages. <https://doi.org/10.1145/3534929>
- [16] Peng Wu, Xiaotao Liu, and Jing Liu. 2023. Weakly Supervised Audio-Visual Violence Detection. *IEEE Trans. Multimedia.* 25 (2023), 1674–1685. <https://doi.org/10.1109/TMM.2022.3147369>
- [17] Robert Gorwa, Reuben Binns, and Christian Katzenbach. 2020. Algorithmic content moderation: Technical and political challenges in the automation of platform governance. *Big Data Soc.* 7, 1 (2020), 1–15. <https://doi.org/10.1177/2053951719897945>
- [18] Raul Gomez, Jaume Gibert, Lluís Gomez, and Dimosthenis Karatzas. 2020. Exploring Hate Speech Detection in Multimodal Publications. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 1470–1478. https://openaccess.thecvf.com/content_WACV_2020/html/Gomez_Exploring_Hate_Speech_Detection_in_Multimodal_Publications_WACV_2020_paper.html
- [19] Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA. 2611–2624.
- [20] Yoonseo Choi, Eun Jeong Kang, Min Kyung Lee, and Juho Kim. 2023. Creator-Friendly Algorithms: Behaviors, Challenges, and Design Opportunities in Algorithmic Platforms. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 564, 22 pages.
- [21] Pantelitsa Leonidou, Nicolas Kourtellis, Nikos Salamanos, and Michael Sirivianos. 2023. Privacy-Preserving Online Content Moderation: A Federated Learning Use Case. In *Companion Proceedings of the ACM Web Conference 2023 (WWW '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 280–289. <https://doi.org/10.1145/3543873>

- [22] Ashlee Milton, Leah Ajmani, Michael Ann DeVito, and Stevie Chancellor. 2023. "I See Me Here": Mental Health Content, Community, and Algorithmic Curation on TikTok. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 480, 1–17. <https://doi.org/10.1145/3544548.3581489>
- [23] Shagun Jhaver, Iris Birman, Eric Gilbert, and Amy Bruckman. 2019. Human-Machine Collaboration for Content Regulation: The Case of Reddit Automoderator. *ACM Trans. Comput.-Hum. Interact.* 26, 5, Article 31 (October 2019), 1–35. <https://doi.org/10.1145/3338243>
- [24] Alon Halevy, Cristian Canton-Ferrer, Hao Ma, Umut Ozertem, Patrick Pantel, Marzieh Saeidi, Fabrizio Silvestri, and Ves Stoyanov. 2022. Preserving integrity in online social networks. *Commun. ACM* 65, 2 (February 2022), 92–98. <https://doi.org/10.1145/3462671>
- [25] Mahi Kolla, Siddharth Salunkhe, Eshwar Chandrasekharan, and Koustuv Saha. 2024. LLM-Mod: Can Large Language Models Assist Content Moderation? In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 217, 1–8. <https://doi.org/10.1145/3613905.3650828>
- [26] Renkai Ma and Yubo Kou. 2021. "How advertiser-friendly is my video?": YouTube's Socioeconomic Interactions with Algorithmic Content Moderation. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 429 (October 2021), 25 pages. <https://doi.org/10.1145/3479573>
- [27] Kelsea Schlenberg, Lingyuan Li, Guo Freeman, Samaneh Zamanifard, and Nathan J. McNeese. 2023. Towards leveraging AI-Based moderation to address emergent harassment in social virtual reality. In *Proceedings of the 2023 CHI conference on human factors in computing systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3544548.3581090> Number of pages: 17 Place: Hamburg, Germany tex.articleno: 514.
- [28] Yaqiong Li, Peng Zhang, Hansu Gu, Tun Lu, Siyuan Qiao, Yubo Shu, Yiyang Shao, and Ning Gu. 2024. DeMod: A Holistic Tool with Explainable Detection and Personalized Modification for Toxicity Censorship. *arXiv preprint arXiv: 2411.01844(2024)*.
- [29] Shagun Jhaver, Himanshu Rathi, and Koustuv Saha. 2024. Bystanders of Online Moderation: Examining the Effects of Witnessing Post-Removal Explanations. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 191, 1–9. <https://doi.org/10.1145/3613904.3642204>
- [30] Mirko Franco, Ombretta Gaggi, and Claudio E. Palazzi. 2023. Analyzing the Use of Large Language Models for Content Moderation with ChatGPT Examples. In *Proceedings of the 3rd International Workshop on Open Challenges in Online Social Networks (OASIS '23)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3599696.3612895>
- [31] Renkai Ma and Yubo Kou. 2022. "I'm not sure what difference is between their content and mine, other than the person itself": A Study of Fairness Perception of Content Moderation on YouTube. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 425 (November 2022), 28 pages. <https://doi.org/10.1145/3555150>
- [32] Shagun Jhaver, Alice Qian Zhang, Quan Ze Chen, Nikhila Natarajan, Ruotong Wang, and Amy X. Zhang. 2023. Personalizing Content Moderation on Social Media: User Perspectives on Moderation Choices, Interface Design, and Labor. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW2, Article 289 (October 2023), 33 pages. <https://doi.org/10.1145/3610080>
- [33] Perna Juneja, Deepika Rama Subramanian, and Tanushree Mitra. 2020. Through the looking glass: Study of transparency in Reddit's moderation practices. *Proceedings of the ACM on Human-Computer Interaction* 4, GROUP (2020), 1–35.
- [34] Katie Canales. 2021. Facebook's AI moderation reportedly can't interpret many languages, leaving users in some countries more susceptible to harmful posts. <https://www.businessinsider.com/facebook-content-moderation-ai-cant-speak-all-languages-2021-9>. Retrieved 19 12 2024.
- [35] Sara Abdali, Sina Shaham, and Bhaskar Krishnamachari. 2024. Multi-modal Misinformation Detection: Approaches, Challenges and Opportunities. *ACM Comput. Surv.* (Oct. 2024). <https://doi.org/10.1145/3697349>
- [36] Bryan Dosono and Bryan Semaan. 2019. Moderation Practices as Emotional Labor in Sustaining Online Communities: The Case of AAPI Identity Work on Reddit. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, Paper 142, 1–13. <https://doi.org/10.1145/3290605.3300372>
- [37] Na Li, Jie Cai, and Donghee Yvette Wohn. 2023. Ignoring As a Moderation Strategy for Volunteer Moderators on Twitch. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. Association for Computing Machinery, New York, NY, USA, Article 169, 1–7. <https://doi.org/10.1145/3544549.3585704>
- [38] Perna Juneja, Deepika Rama Subramanian, and Tanushree Mitra. 2020. Through the looking glass: Study of transparency in Reddit's moderation practices. *Proceedings of the ACM on Human-Computer Interaction* 4, GROUP (2020), 1–35.
- [39] Eshwar Chandrasekharan, Chaitrali Gandhi, Matthew Wortley Mustelier, and Eric Gilbert. 2019. Crossmod: A Cross-Community Learning-based System to Assist Reddit Moderators. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 174 (November 2019), 30 pages. <https://doi.org/10.1145/3359276>
- [40] Sarah Myers West. 2018. Censored, suspended, shadowbanned: User interpretations of content moderation on social media platforms. *New Media & Society* 20, 11 (2018), 4366–4383. <https://doi.org/10.1177/1461444818773059>
- [41] Piera Riccio, Thomas Hofmann, and Nuria Oliver. 2024. Exposed or Erased: Algorithmic Censorship of Nudity in Art. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 26, 1–17. <https://doi.org/10.1145/3613904.3642586>
- [42] Katrin Hartwig, Tom Biselli, Franziska Schneider, and Christian Reuter. 2024. From Adolescents' Eyes: Assessing an Indicator-Based Intervention to Combat Misinformation on TikTok. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 905, 1–20. <https://doi.org/10.1145/3613904.364226>
- [43] Danding Wang, Qian Yang, Ashraf Abdul, and Brian Y Lim. 2019. Designing theory-driven user-centric explainable AI. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. ACM, Glasgow, UK, 1–15.
- [44] Gordon Pennycook, Jonathon McPhetres, Yunhao Zhang, Jackson G. Lu, and David G. Rand. 2020. Fighting COVID-19 Misinformation on Social Media: Experimental Evidence for a Scalable Accuracy-Nudge Intervention. *Psychol. Sci.* 31, 7 (2020), 770–780. <https://doi.org/10.1177/0956797620939054>
- [45] Johanna Brewer, Morgan Romine, and T. L. Taylor. 2020. Inclusion at Scale: Deploying a Community-Driven Moderation Intervention on Twitch. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 757–769. <https://doi.org/10.1145/3357236.3395514>
- [46] Farnaz Jahanbakhsh, Amy X. Zhang, Adam J. Berinsky, Gordon Pennycook, David G. Rand, and David R. Karger. 2021. Exploring Lightweight Interventions at Posting Time to Reduce the Sharing of Misinformation on Social Media. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 18 (April 2021), 42 pages. <https://doi.org/10.1145/3449092>
- [47] Kristen Vaccaro, Ziang Xiao, Kevin Hamilton, and Karrie Karahalios. 2021. Contestability For Content Moderation. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 318 (oct 2021), 28 pages. <https://doi.org/10.1145/3476059>
- [48] Dokyun Lee, Sangeun Seo, Chanwoo Park, Sunjun Kim, Buru Chang, and Jean Y Song. 2024. Exploring Intervention Techniques to Alleviate Negative Emotions during Video Content Moderation Tasks as a Worker-centered Task Design. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*. Association for Computing Machinery, New York, NY, USA, 1701–1721. <https://doi.org/10.1145/3643834.3660708>
- [49] Han Wang, Tan Rui Yang, Usman Naseem, and Roy Ka-Wei Lee. 2024. Multi-HateClip: A Multilingual Benchmark Dataset for Hateful Video Detection on YouTube and Bilibili. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*. Association for Computing Machinery, New York, NY, USA, 7493–7502. <https://doi.org/10.1145/3664647.3681521>
- [50] Stefano Cresci, Amaury Trujillo, and Tiziano Fagnì. 2022. Personalized Interventions for Online Moderation. In *Proceedings of the 33rd ACM Conference on Hypertext and Social Media (HT '22)*. Association for Computing Machinery, New York, NY, USA, 248–251. <https://doi.org/10.1145/3511095.3536369>
- [51] Deepak Kumar, Yousef AbuHashem, and Zakir Durumeric. 2023. Watch Your Language: Large Language Models and Content Moderation. *arXiv preprint arXiv:2309.14517 (2023)*.
- [52] Todor Markov, Chong Zhang, Sandhini Agarwal, Florentine Eloundou Nekoul, Theodore Lee, Steven Adler, Angela Jiang, and Lilian Weng. 2023. A holistic approach to undesired content detection in the real world. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 15009–15018.
- [53] Jiang Zhang, Qiong Wu, Yiming Xu, Cheng Cao, Zheng Du, and Konstantinos Psounis. 2024. Efficient Toxic Content Detection by Bootstrapping and Distilling Large Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 38, 19 (2024), 21779–21787. <https://doi.org/10.1609/aaai.v38i19.30178>
- [54] Katie Canales. 2021. Facebook's AI moderation reportedly can't interpret many languages, leaving users in some countries more susceptible to harmful posts. <https://www.businessinsider.com/facebook-content-moderation-ai-cant-speak-all-languages-2021-9>. Retrieved 19122024.
- [55] Jan Kirchner and Christian Reuter. 2020. Countering Fake News: A Comparison of Possible Solutions Regarding User Acceptance and Effectiveness. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 140 (October 2020), 27 pages. <https://doi.org/10.1145/3415211>
- [56] Farnaz Jahanbakhsh and David R Karger. 2024. A Browser Extension for in-place Signaling and Assessment of Misinformation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 946, 1–21. <https://doi.org/10.1145/3613904.3642473>
- [57] Yao Lyu, Jie Cai, Anisa Callis, Kelley Cotter, and John M. Carroll. 2024. "I Got Flagged for Supposed Bullying, Even Though It Was in Response to Someone Harassing Me About My Disability": A Study of Blind TikTok's Content Moderation Experiences. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 741, 1–15. <https://doi.org/10.1145/3613904.3642148>

- [58] Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep Reinforcement Learning from Human Preferences. In Proceedings of the 31st International Conference on Neural Information Processing Systems, Long Beach, California, USA, 4302 - 4310. https://proceedings.neurips.cc/paper_files/paper/2017/file/d5e2c0adad503c91f91df240d0cd4e49-Paper.pdf
- [59] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-VL: A Versatile Vision-Language Model for Understanding, Localization, Text Reading, and Beyond. arXiv:2308.12966 [cs.CV] <https://arxiv.org/abs/2308.12966>
- [60] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, Guanting Dong, Haoran Wei, Huan Lin, Jialong Tang, Jialin Wang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Ma, Jianxin Yang, Jin Xu, Jingren Zhou, Jinze Bai, Jinzheng He, Junyang Lin, Kai Dang, Keming Lu, Keqin Chen, Kexin Yang, Mei Li, Mingfeng Xue, Na Ni, Pei Zhang, Peng Wang, Ru Peng, Rui Men, Ruize Gao, Runji Lin, Shijie Wang, Shuai Bai, Sinan Tan, Tianhao Li, Tianyu Liu, Wenbin Ge, Xiaodong Deng, Xiaohuan Zhou, Xingzhang Ren, Xinyu Zhang, Xipin Wei, Xuancheng Ren, Xuejing Liu, Yang Fan, Yang Yao, Yichang Zhang, Yu Wan, Yunfei Chu, Yuyang Liu, Zeyu Cui, Zhenru Zhang, Zhifang Guo, and Zhihao Fan. 2024. Qwen2 Technical Report. arXiv:2407.10671 [cs.CL] <https://arxiv.org/abs/2407.10671>
- [61] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 770–778. https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html
- [62] Gunnar Farneback. 2003. Two-Frame Motion Estimation Based on Polynomial Expansion. In Image Analysis, Josef Bigun and Tomas Gustavsson (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 363–370.
- [63] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine Mcleavey, and Ilya Sutskever. 2023. Robust Speech Recognition via Large-Scale Weak Supervision. In Proceedings of the 40th International Conference on Machine Learning, Honolulu, Hawaii, USA, 28492–28518. <https://dl.acm.org/doi/abs/10.5555/3618408.3619590>
- [64] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes and Yejin Choi. 2020. The Curious Case of Neural Text Degeneration. arXiv: 1904.09751 [cs.CL] <https://arxiv.org/abs/1904.09751>
- [65] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention Is All You Need. In Proceedings of the 31st International Conference on Neural Information Processing Systems, Long Beach, California, USA. https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

A APPENDICES

A.1 Rating Standards of Responsibility Dimensions

The following content outlines the connotations of the responsibility dimensions. The rating standards for each dimension are shown in Table 1, which are embedded in *ResponSight* to guide the model in video content rating.

“Compliance”: This dimension includes verifying legal compliance by checking for content that involves violence, explicit material, or infringement of others’ rights. Compliance also covers adherence to platform policies, monitoring for elements like hate speech, harassment, or deceptive advertising, and examining intellectual property protections.

“Reliability”: This dimension evaluates the sources of information used in the video, emphasizing transparency by confirming that reliable sources are cited, and any affiliations, sponsorships, or personal biases are disclosed. Additionally, reliability assesses data support and objectivity, ensuring that all claims are evidence-based and presented neutrally. It also considers the integrity of video editing, checking that manipulative techniques or deepfake technologies have not been used in ways that could distort viewers’ understanding of the content.

“Fairness and Inclusivity”: This dimension reviews content to ensure it avoids harmful references to race, gender, religion, sexual orientation, or social class, while also examining diversity in representation to confirm that varied cultural and social perspectives are respectfully included. It is also encouraged to inspire open discussion, supporting the inclusion of diverse viewpoints without suppressing differing expressions.

“Privacy and Data Security”: This dimension checks for the protection of private information, such as addresses, phone numbers, and identification numbers, verifying that these details are adequately safeguarded. For content involving sensitive data like health, financial, or confidential business information, it assesses whether clear privacy statements are included to protect the rights and privacy of all individuals involved.

“Social Impact and Responsibility”: This dimension assesses the potential for content to provoke harmful social attitudes, spread divisive messages, or cause adverse reactions through provocative elements, particularly when such content risks influencing public opinion or social behavior in a detrimental way.

“Appropriateness of Emotional Expression”: This dimension evaluates if the video uses heightened language, intense visuals, or exaggerated emotional cues to provoke strong responses such as fear, anger, or anxiety in viewers. Excessive emotional expression, particularly when it risks stirring broad negative sentiment or distress, is carefully reviewed to ensure that content is balanced and respects the viewer’s emotional state.

A.2 Explainable Evaluation Module

A.2.1 Video Decomposer. Initially, the uploaded video is processed by the video decomposer, which separates the visual and audio elements, extracts keyframes, and recognizes audio content to provide essential inputs for subsequent multimodal analysis.

The first step involves capturing a series of image frames at fixed intervals (one frame per second). To refine this frame extraction, we employ a combination of the ResNet-50 model and optical flow analysis to extract keyframes effectively. ResNet is a convolutional neural network with strong feature extraction capabilities, that effectively captures fine-grained details and distinguishes key visual shifts across scenes [61], which can identify frames that reflect meaningful transitions or events within the video. In this decomposer, we apply ResNet-50 with 50 layers deep, balancing model performance and computing resource consumption. With the short-cut connections in residual blocks, the model is able to keep the original information to prevent overfitting or gradient issues. Each building block was defined as:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (1)$$

where x is the input of vectors, $\mathcal{F}(x, \{W_i\})$ shows the residual mapping to be learned, y is the output integrated with the residual and the input vector.

In parallel, optical flow analysis is utilized to capture motion dynamics between consecutive frames, by calculating the motion vectors of pixels between frames to detect significant changes in scene dynamics or object movement [62]. The integration combines the strengths of both approaches and enables the system to capture a diverse range of significant frames, balancing content detail with scene transitions. To manage the number of extracted keyframes

Table 1: Rating standards of responsibility dimensions

Responsibility Dimensions	Rating Standards
Compliance	A: Fully comply with laws and platform rules, with no violations. B: Generally compliant, with minor infractions that do not affect overall legality. C: Insufficient compliance information, unclear if fully legal. D: Contains clear violations of laws or platform rules.
Reliability	A: All sources are authoritative and transparent, with reliable data and no signs of manipulation. B: Sources are mostly reliable, with accurate data, though some details may be lacking. C: Some sources or data are missing, making it hard to fully assess reliability. D: Sources are unclear, data is insufficient, and manipulation is apparent.
Fairness & Inclusivity	A: No discriminatory content, fully represents diverse viewpoints and cultural backgrounds. B: Mostly fair, with some diversity shown but minor gaps. C: No obvious discrimination, but the information is incomplete, making it hard to assess diversity. D: Clearly discriminatory content, lacks diversity, and contains bias.
Privacy & Data Security	A: Strong privacy protection, with no unauthorized disclosure of personal information. B: Generally protects privacy, but with minor unauthorized disclosures, though no significant risks. C: Privacy policies or information are unclear, and some privacy risks may be present. D: Serious privacy violations, including the disclosure of sensitive personal information.
Social Impact & Responsibility	A: The content has a clear positive impact on society, encouraging productive discussion. B: Generally positive, but with limited influence or shallow discussion. C: No clear positive or negative impact, with some missing information making it hard to assess. D: Clear negative impact, spreading misinformation or causing harm.
Appropriateness of Emotional Expression	A: Emotional expression is natural and appropriate, with no manipulation or exaggeration. B: Emotional expression is mostly appropriate, with minor exaggeration or emotional triggers. C: Neutral emotional expression, with insufficient information to determine manipulation. D: Strong emotional manipulation or exaggeration, intended to evoke unnecessary emotional responses.

and enhance processing efficiency, we apply an automated threshold adjustment method by setting the threshold at the upper quartile of visual and motion-based differences, this approach ensures that only frames with distinctive visual and motion changes are retained while accommodating variations across different video types.

Audio recognition is processed through Whisper model for speech-to-text conversion, with high-performance extracting transcripts in diverse video content [63]. This transcript serves as a textual representation of the video’s auditory content, providing a comprehensive foundation for subsequent analyses.

A.2.2 Prompt Generator. Once the transcript is recorded, it is passed to the prompt generator, a module designed to tailor prompts specifically to the video’s context, enhancing adaptability to diverse content types and ensuring relevance across various themes and styles. This module employs prompt chaining, decomposing the primary task into multiple subtasks, with targeted prompts generated for each. The process encompasses three main components: content awareness, monitoring point focus, and adaptive prompt generation.

First, the transcript is inputted into a Large Language Model (LLM) to generate a high-level summary that captures the video’s primary themes and topics, which enables the model to contextualize the video’s genre and main subject matter. Then the model is required to identify critical monitoring points by referencing the six dimensions of responsibility, explaining why certain areas might present risks. This approach allows the system to focus precisely on specific areas that warrant closer examination. Finally, based

on these key monitoring points, the model generates an adaptive prompt that guides the subsequent model in detailed content analysis. With the instruction component, the prompt can guide the model to produce structured, actionable output, ensuring clarity and relevance in the analytical results.

A.2.3 MLLM-based Detector. Following prompt generation, the detector analyses the content using MLLMs that interpret both visual and textual elements, assessing content against predefined social responsibility criteria. This detector delivers insights into complex content, helping to capture implicit and explicit signals of responsible or irresponsible elements.

In frame analysis, the model first processes all keyframes individually, identifying and analyzing core elements within each frame such as environment, people, emotions, body language, interactions, on-screen text, and other prominent visual aspects. For example, the model should identify misleading visuals, and inappropriate imagery. Based on this, the model assesses potentially sensitive or risk-related aspects based on specific responsibility dimensions. Results from all keyframes are synthesized to form a comprehensive evaluation across the entire video, with the adaptive prompt guiding the model to assess overarching patterns and content risks. In parallel, for transcript analysis, the model uses the adaptive prompt to evaluate the transcript holistically, focusing on areas such as dialogue, scenes, actions, and narrative progression that may raise ethical or responsibility concerns. To analyze the materials, the model transforms text into high-dimensional vectors to capture semantic and contextual information. It employs

probabilistic sampling to predict the next word based on previous words $u = [u_1, u_2, \dots, u_t]$, modelling the probability distribution [64]. Probability can be calculated by:

$$P(u_i | u_{<i}) = \frac{\exp(l_i/t)}{\sum_{i'} \exp(l_{i'}/t)} \quad (2)$$

where l_i is the intermediate value before probability conversion, t is the temperature parameter. To ensure output stability and reliability, temperature sampling with a low temperature parameter reduces randomness.

Moreover, attention mechanisms within transformer architectures enhance the identification of critical ethical content by concentrating on salient words or sentences, thereby improving the accuracy of textual analysis [65]. In image processing, models such as Vision Transformers segment images into small patches, treating these as sequences of visual tokens like word tokens in text processing. This approach enables the capture of long-range dependencies and global contextual information within the image. Attention mechanisms are then employed to focus on important regions pertinent to the specific task, further enhancing analytical precision.

After single-modality analysis, the detector generates textual analyses from images and transcripts, denoted as S_{frame} and $S_{transcript}$, respectively. These analyses are fused by $F: S_{frame} \times S_{transcript}$, aligning the frame and transcript evaluations based on the six responsibility dimensions to achieve a cohesive, multimodal assessment. This fusion ensures temporal and semantic alignment, enabling the model to construct a unified understanding of the content. Consequently, it allows for a more accurate identification of cross-modal patterns and interactions. This dimension-focused integration provides a comprehensive and precise evaluation of the video, reinforcing a robust foundation for responsible content moderation.

All analysis results are synthesized by the rating function, which aggregates and quantifies the evaluation into a comprehensive rating score, complemented by an in-depth evaluation report, offering creators a clear, explainable outcome and actionable insights to enhance the responsibility level of their content. We leverage a prompting chain approach, dividing the rating task into four stages for accurate and systematic evaluation.

In the initial stage, video content, type, single-modality recognition results, and multimodal analysis outputs are integrated into a unified input for the model, ensuring completeness of materials for accurate rating. The aggregated information is processed by the model's memory system, which establishes a foundational context for rating across all dimensions.

Next, the model is guided by a set of rating standards designed for six responsibility dimensions. The model is required to output specific reasons and problem elements identified within the content along with the rating. In this rating process, we start with the original text material T and the rating standards S . The large model performs n independent evaluations using a rating function r , incorporating random variables θ_i to introduce variability:

$$R_i = r(T, S, \theta_i), \text{ for } i = 1, 2, \dots, n \quad (3)$$

Each R_i represents a rating grade from the set $\{A, B, C, D\}$.

Based on the sub-dimension scores, the final rating for the video is determined. We employed a Chain-of-Thought (CoT) approach

in designing the rating prompt, guiding the model to follow a structured decision-making process and execute each step systematically to reach an accurate conclusion. This method requires the model to follow explicit logic and procedural steps to arrive at accurate conclusions. First, a basic rating logic is established: if the Compliance dimension receives a D rating, the model is instructed to bypass all further calculations and immediately output a D for the overall rating, preventing potential errors in calculation logic that could misrepresent critical compliance failures. Then, the model needs to map each sub-dimension rating to numerical scores. We map each rating R_i to a numerical value V_i using a function f :

$$V_i = f(R_i) = \begin{cases} v_A, & \text{if } R_i = A \\ v_B, & \text{if } R_i = B \\ v_C, & \text{if } R_i = C \\ v_D, & \text{if } R_i = D \end{cases} \quad (4)$$

In our setting, $v = \{v_A = 3, v_B = 2, v_C = 1, v_D = 0\}$. The model is then guided through detailed steps for calculating an average score, including summation and division procedures. This structured prompting is intended to ensure that the model correctly interprets each calculation step, thereby minimizing the risk of computational inaccuracies. The average numerical score is then calculated by the arithmetic mean. Once the average score is derived, we map the average score $\bar{V}$ to the final rating L using a piecewise function g :

$$L = g(\bar{V}) = \begin{cases} A, & \text{if } \tau_1 \leq \bar{V} \leq v_A \\ B, & \text{if } \tau_2 \leq \bar{V} < \tau_1 \\ C, & \text{if } \tau_3 \leq \bar{V} < \tau_2 \\ D, & \text{if } v_D \leq \bar{V} < \tau_3 \end{cases} \quad (5)$$

where τ_1, τ_2, τ_3 are predefined thresholds determining the grade boundaries. In our setting, $\tau = \{\tau_1 = 2.25, \tau_2 = 1.5, \tau_3 = 0.75\}$. This approach integrates multiple assessments, quantifies them through numerical mapping, and aggregates the results to produce a consistent and objective final rating aligned with the original materials and rating standards. The output includes an evaluation report that provides a general commentary on the video, the assigned responsibility rating, and specific issues observed within image and transcript elements.

A.3 Adaptive Suggestion Module

The function interprets the evaluation findings to generate video-specific recommendations that align with responsibility goals driven by LLM. This structured approach prevents disjointed advice by linking each recommendation directly to specific content goals or issues identified in the evaluation. The model generates suggestions according to the video's rating level. For high-rated videos (Grade A), it focuses on visibility and popularity enhancement, such as refining titles, descriptions, and keywords to increase reach. For videos rated B or C, the advice emphasizes responsible content improvements, including refining factual accuracy, enhancing transparency, and adding clarifications where necessary. For lower-rated videos (Grade D), substantial revisions are recommended to address core issues, providing clear guidance for correcting misleading or inaccurate elements.

Beyond general content-level advice, the model offers specific suggestions for visual and textual components. Image-based recommendations may involve adjusting visual elements, updating scene

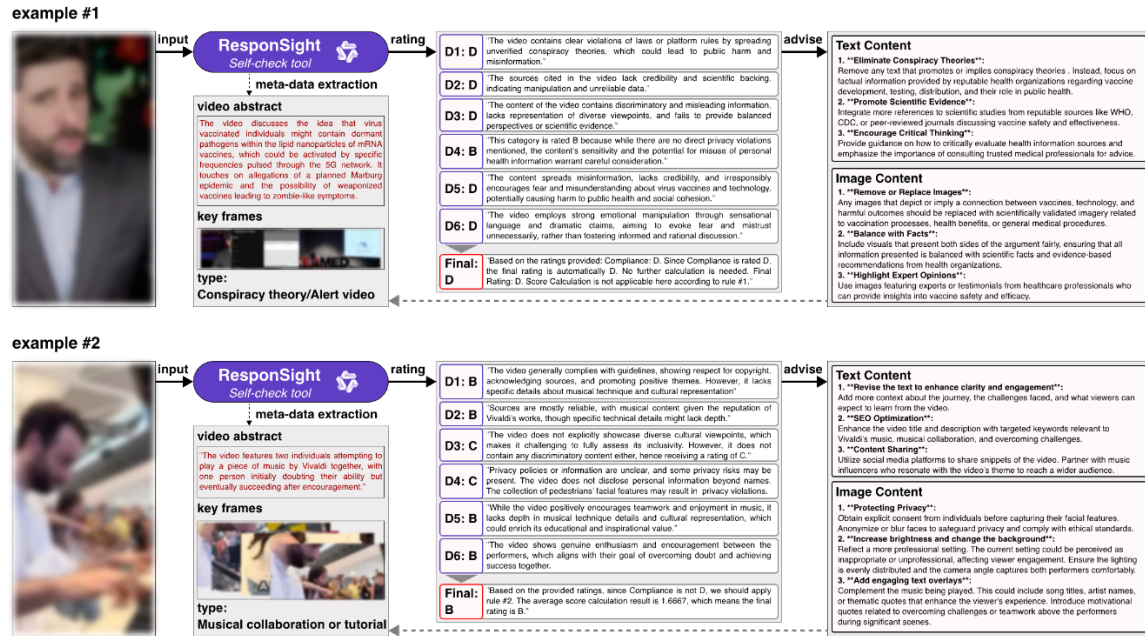


Figure 5: Examples of the video rating results

settings, or replacing inappropriate images, while text-based guidance may focus on enriching content depth, adjusting emotional tone, or improving logical accuracy. Each recommendation includes concrete examples to facilitate implementation, ensuring that creators can make precise, effective improvements in alignment with platform expectations. This suggestion can empower creators to align with platform standards more effectively, promoting content that resonates responsibly with audiences.

A.4 Video Evaluation Examples

We conducted manual testing by inputting platform-scraped video samples into *ResponSight* for automated rating and recommendation generation. Figure 5 showcases two video content evaluation cases.